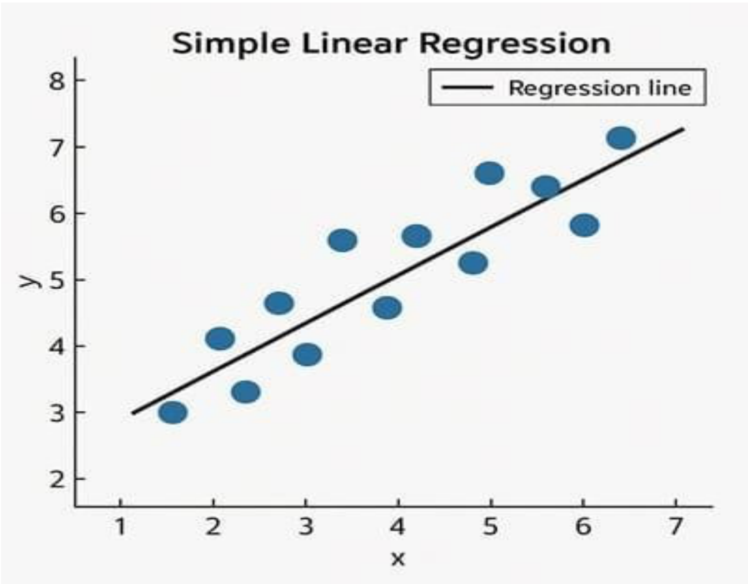
Algorithms

1. Linear Regression

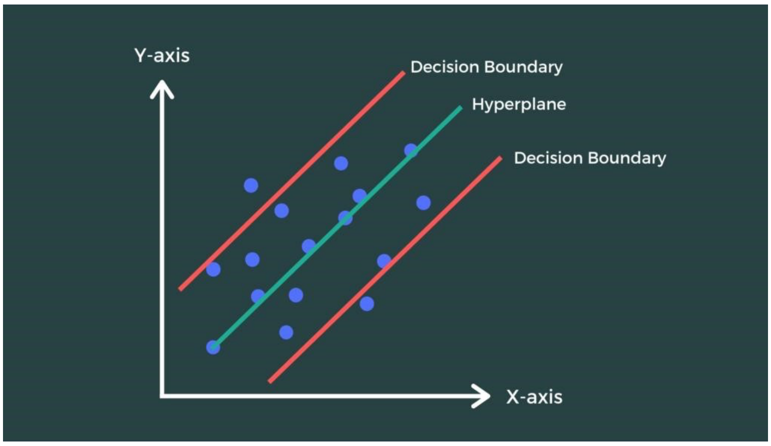
Descrip1on:

Linear Regression is a simple supervised learning algorithm used for predic6ng con6nuous values. It finds a straight-line rela6onship between input features and output. In house price predic6on, it es6mates price based on features like bedrooms, sq@, etc. It is fast and easy but works best when data has a linear paBern.



Support Vector Regression (SVR)

Descrip1on:

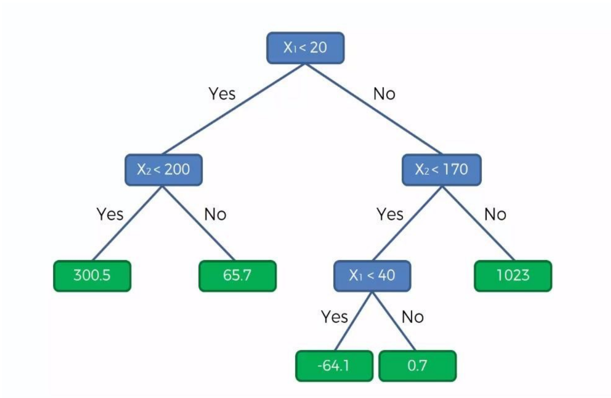


SVR is the regression version of Support Vector Machine. It tries to fit the best line within a margin (called epsilon tube) while minimizing error. It is powerful for complex data but sensi6ve to scaling. In this project, it helps model non-linear rela6onships between house features and price.

Decision Tree Regressor

Descrip1on:

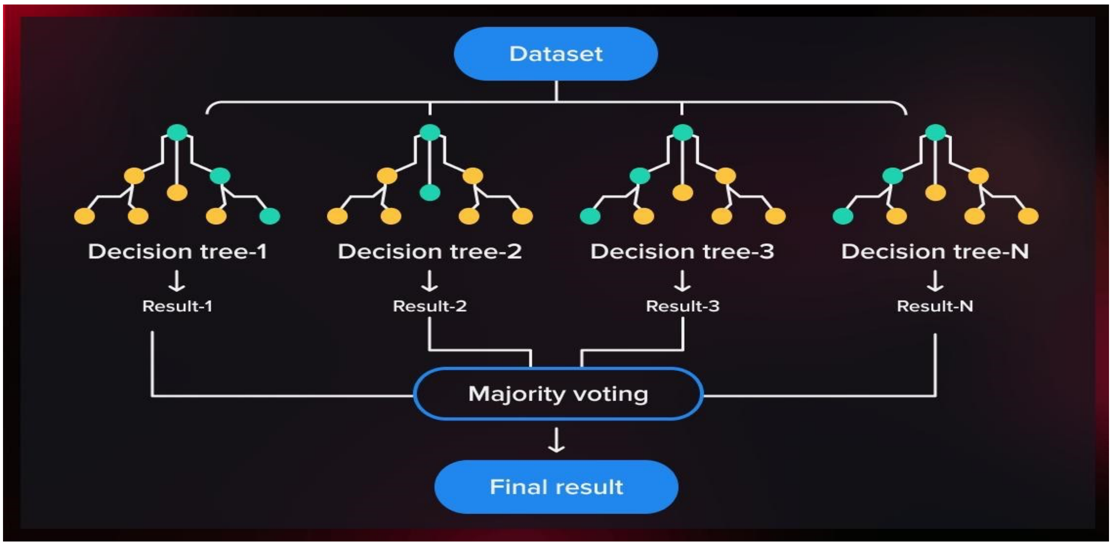
A Decision Tree Regressor splits data into branches based on feature values. Each split reduces error and leads to a final predic6on at the leaf node. It is easy to understand but may overfit if the tree becomes too deep.



Random Forest Regressor

Descrip1on:

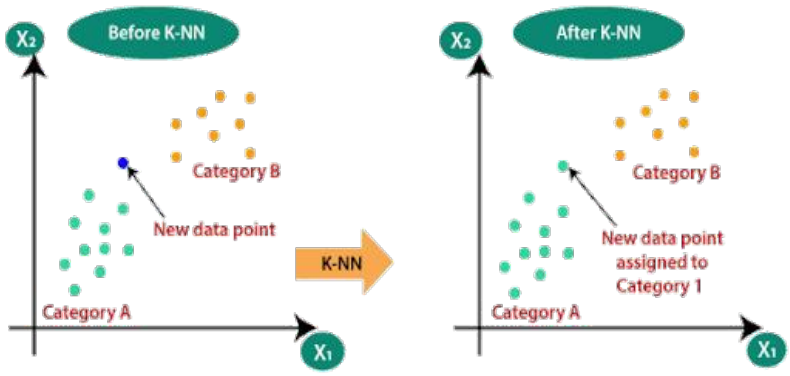
Random Forest is an ensemble method that combines mul6ple decision trees. Each tree gives a predic6on, and the final output is the average of all. It reduces overfiPng and improves accuracy, making it one of the best models for this dataset.



K-Nearest Neighbors (KNN) Regressor

Descrip1on:

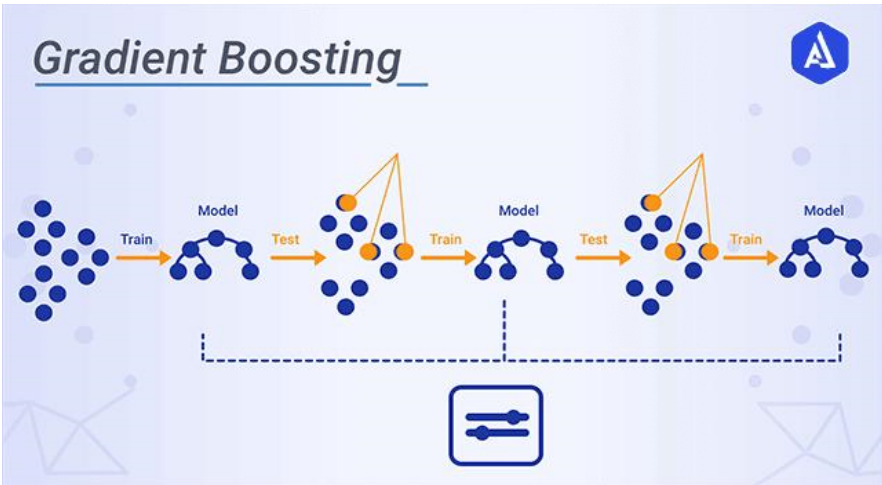
KNN predicts house price based on the average of nearest similar houses. It uses distance (like Euclidean distance) to find similar data points. It is simple but slower for large datasets.



2. Gradient Boos1ng Regressor

Descrip1on:

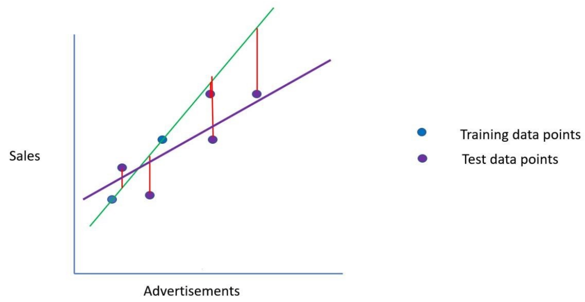
Gradient Boos6ng builds models sequen6ally, where each new model corrects errors of the previous one. It gives very high accuracy and is widely used in real-world applica6ons. It performs very well in house price predic6on.



3. Ridge Regression

Descrip1on:

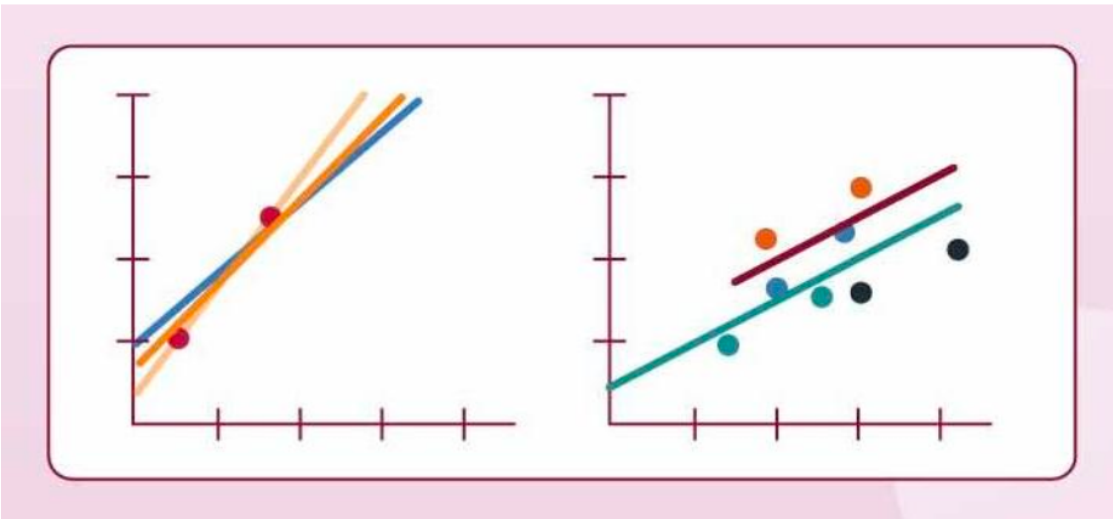
Ridge Regression is an improved version of Linear Regression. It adds a penalty (regulariza6on) to reduce overfiPng. It works well when features are correlated and improves model stability.



8. Lasso Regression

Description:

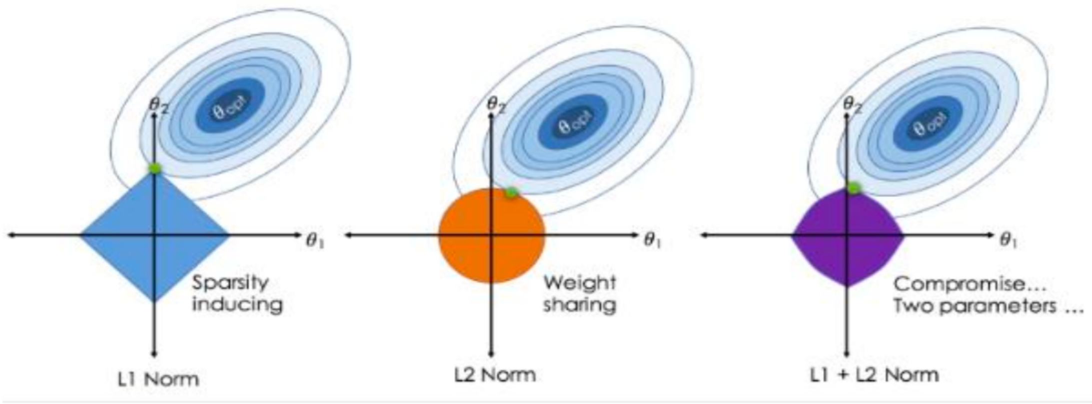
A linear model with L1 regularization that can shrink some coefficients to zero. Effectively performs feature selection. Helps simplify the model.



8. Lasso Regression

Description:

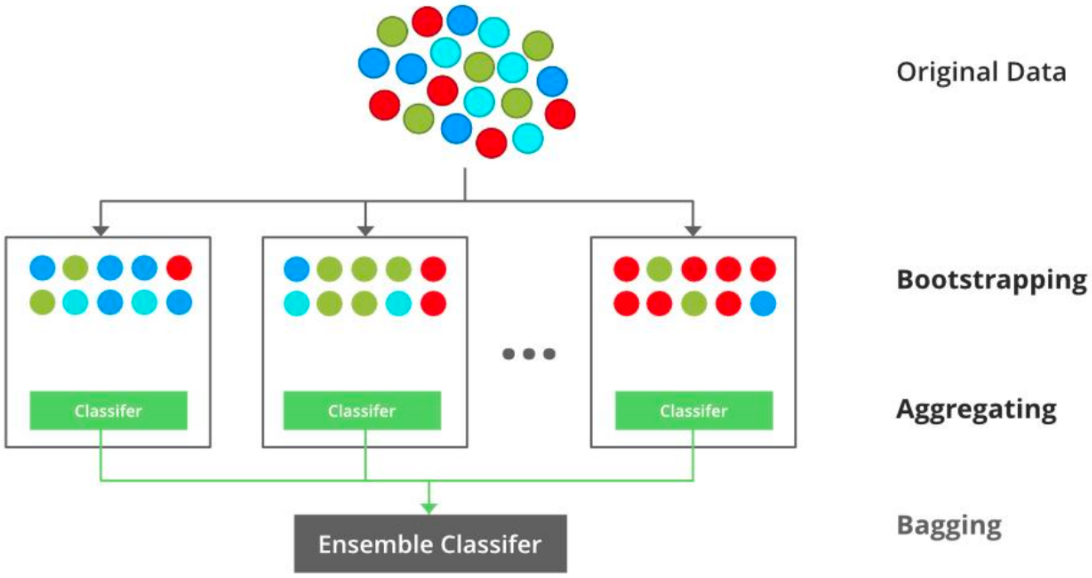
A linear model with L1 regularization that can shrink some coefficients to zero. Effectively performs feature selection. Helps simplify the model.



10. XGBoost Regression

Description:

An advanced boosting algorithm that builds trees sequentially with optimization techniques. Handles missing values and complex patterns efficiently. Widely used in competitions.



Final Conclusion:

This project was done to predict house prices using different machine learning algorithms and house details like bedrooms, bathrooms, and square footage. Several models were tested to see which one gives more accurate predic6ons. Simple models like Linear Regression work well for basic predic6ons, while advanced models like Random Forest and Gradient Boos6ng give beBer results by learning paBerns from the data more effec6vely. These advanced models also help reduce predic6on errors. Overall, the project shows that ensemble models are more reliable for predic6ng house prices accurately.

What I Learned:

- I learned how machine learning can help in predic6ng house prices using house details.
- I understood how different regression models work and how their results can change.
- I learned why preparing the data properly is important before training a model.
- I understood how to check which model performs beBer using accuracy measures.
- I learned that models like Random Forest and Gradient Boos6ng usually give beBer predic6ons.
- Overall, this project gave me prac6cal experience in building and tes6ng machine learning models.